

An Explainable Fake News Detection Using Classical Machine Learning

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Abstract—Fake news detection has become a critical challenge in the digital era due to the rapid dissemination of misinformation through online platforms. This study presents an explainable fake news detection framework based on classical machine learning techniques using the LIAR dataset. Initially, baseline models including Logistic Regression (LR) and Support Vector Machines (SVM) are reproduced using TF-IDF feature representations to establish a reference performance. To address the limitations of text-only models, the proposed approach incorporates enhanced feature engineering by combining textual features with meta-data attributes and applying feature selection techniques such as Chi-square (χ^2) and Mutual Information (MI) to reduce dimensionality and improve discriminative power. Furthermore, additional models including Naive Bayes (NB), Random Forest (RF), and Extreme Gradient Boosting (XGBoost) are evaluated with hyperparameter optimization. Experimental results demonstrate that ensemble-based models, particularly XGBoost, achieve superior performance with an F1-score of up to 0.87, outperforming baseline models. The findings also highlight that feature selection plays a more significant role than model choice, with Mutual Information providing more robust performance across models. To enhance transparency, explainable artificial intelligence techniques, specifically SHAP, are employed to interpret both global feature importance and individual predictions. The analysis reveals that a small set of discriminative linguistic features significantly influences model decisions, validating the effectiveness of the proposed feature selection strategy. In-general, this study demonstrates that integrating feature engineering, model optimization, and explainability can significantly improve both the performance and interpretability of fake news detection systems.

I. INTRODUCTION

The rapid growth of online media platforms has significantly increased the spread of misinformation and fake news, posing serious risks to individuals, organizations, and democratic processes [1]. Fake news, defined as deliberately misleading or false information presented as legitimate news, has the potential to influence public perception and decision-making on a large scale [2]. As a result, the development of automated methods to detect fake news has become a crucial research area in machine learning and data science [3].

Traditional approaches to fake news detection treat the problem as a text classification task, where machine learning models are trained to distinguish between true and deceptive content based on linguistic patterns [4]. Previous studies have demonstrated the effectiveness of classical machine learning algorithms such as LR and SVM, along with ensemble-based

methods including RF and XGB, when combined with text-based feature extraction techniques such as TF-IDF [5], which captures the importance of the term relative to the corpus. In particular, the LIAR dataset introduced by Wang (2017) has become a standard benchmark for evaluating fake news detection models due to its diverse and challenging set of labeled political statements [6].

Despite the success of these models, one major limitation is the lack of interpretability. Most existing approaches focus primarily on improving classification accuracy, often neglecting the need to understand why a model predicts a news statement as fake or real [3]. This lack of transparency can reduce trust in automated systems, especially in sensitive domains such as political information analysis [7].

To address this limitation, this study aims to develop an explainable fake news detection framework using classical machine learning techniques. First, baseline models are reproduced following established methodologies to ensure consistency with previous research [6]. Subsequently, improvements are introduced through feature engineering and model optimization. The key contribution of this work lies in integrating explainable artificial intelligence techniques to interpret model predictions and identify the most influential features driving classification decisions.

The remainder of this paper is organized as follows: Section II reviews related work in fake news detection, Section III describes the dataset used, Section IV presents the methodology, Section V is baseline reproduction, Section VI discusses the proposed improvements, Section VII is results and discussion, Section VIII presents the explainability analysis, and Section IX concludes the study with future research directions.

II. RELATED WORK

Fake news detection has become a critical research area in recent years due to the rapid spread of misinformation through online platforms. Early approaches primarily focused on treating this problem as a text classification problem [1], [3]. One of the most widely used benchmark datasets in this domain is the LIAR dataset introduced by Wang (2017) [6]. This dataset consists of 12,836 short political statements labeled across six fine-grained truthfulness categories. Wang (2017) evaluated several classical machine learning models, including Logistic Regression and Support Vector Machines,

alongside deep learning approaches such as Convolutional Neural Networks. According to the results, traditional machine learning models achieve moderate performance.

Shu et al. (2017) [3] presented a comprehensive overview of fake news detection from a data mining perspective, emphasizing the unique challenges associated with this problem. The authors proposed a unified framework consisting of feature extraction and model construction as main components. The features include news content features and social context features. In this study, a similar implementation strategy is adopted, where TF-IDF features are combined with supervised learning algorithms such as Logistic Regression, Support Vector Machines, and ensemble methods to perform classification tasks [9].

Despite these advancements, a significant limitation is the lack of interpretability. Most approaches focus on maximizing predictive accuracy, while providing limited insight into model decision-making processes. Therefore, this study reproduces baseline models evaluated by Wang (2017) [6], specifically Logistic Regression and Support Vector Machines, using TF-IDF feature representations. Additionally, further improvements are introduced by incorporating Naïve Bayes, Random Forest, and XGBoost models along with Chi-square and Mutual Information feature selection techniques to enhance classification performance.

Furthermore, this study addresses the limitation of interpretability by integrating explainable artificial intelligence techniques to provide more transparent and insightful predictions.

III. DATASET DESCRIPTION

In this study, the LIAR dataset, introduced by Wang (2017) [6], is utilized as a benchmark dataset for fake news detection. The dataset is publicly available and can be accessed via Kaggle [16]. The dataset is collected from the PolitiFact fact-checking platform and consists of 12,836 short political statements annotated with fine-grained truthfulness labels. These annotations are provided by professional editors, ensuring high data quality, with an inter-annotator agreement of 0.82 measured using Cohen's kappa [6].

The dataset includes six truthfulness categories: pants-fire, false, barely-true, half-true, mostly-true, and true [6]. The class distribution is relatively balanced, with approximately 1,050 instances in the pants-fire category and between 2,063 and 2,638 instances for each of the remaining classes.

For the purpose of this study, the original multi-class problem is transformed into a binary classification task. Specifically, the labels pants-fire, false, and barely-true are grouped as fake, while half-true, mostly-true, and true are grouped as real. This transformation simplifies the classification problem and facilitates comparison with traditional binary fake news detection approaches, while still leveraging the semantic richness of the original labels.

In addition to textual statements, the dataset provides several meta-data attributes, including speaker identity, party affiliation, context, and job title. However, this study focuses

primarily on textual features derived from the statement content to maintain consistency with classical machine learning approaches.

The LIAR dataset is pre-partitioned into training, validation, and test sets, enabling reproducibility and fair comparison with prior work. Its real-world nature, balanced class distribution, and widespread adoption in the literature make it well-suited for this research.

Despite these advantages, the dataset has certain limitations. The statements are relatively short and lack broader contextual information, which increases the difficulty of classification. Furthermore, the dataset is limited to political content, which may affect the generalizability of models to other domains. These challenges highlight the importance of robust feature extraction and model selection techniques for effective fake news detection.

IV. METHODOLOGY

This study follows a structured machine learning pipeline for fake news detection, consisting of data preprocessing, feature extraction, model training, and evaluation. The methodology is designed to reproduce baseline models from Wang (2017) [6] and extend them with additional machine learning techniques to improve performance and interpretability.

A. Data Preprocessing

The LIAR dataset consists of short textual statements that are often noisy and unstructured. There are several preprocessing steps applied to clean and standardize the text data.

- Converting text to lowercase
- Removing punctuation and non-alphabetic characters
- Removing stop words
- Tokenization and Stemming

These preprocessing techniques help reduce noise and improve the quality of feature representation, which is essential for effective model training.

In addition, the original six-class labels in the LIAR dataset are converted into a binary classification setting. The labels pants-fire, false, and barely-true are grouped as “fake”, while half-true, mostly-true, and true are grouped as “real”. This transformation simplifies the classification task and aligns with practical fake news detection scenarios.

B. Feature Representation

1) *Text Features*: The textual statements are transformed into numerical representations using TF-IDF vectorization [5]. This approach captures the importance of words relative to the corpus while reducing the influence of common terms. Unigram and bigram features are used to incorporate limited contextual information.

2) *Meta-data Features (Proposed Extension)*: In addition to textual content, the LIAR dataset provides several meta-data attributes, including:

- Speaker
- Subject
- Party affiliation

- Context
- Job
- State

Unlike the baseline model, which relies solely on text, the proposed approach integrates these additional features. Categorical attributes are encoded using one-hot encoding and combined with TF-IDF features to form a unified feature space.

C. Baseline Models

To reproduce the baseline results reported by Wang (2017) [6], the following classical machine learning models are implemented:

- Logistic Regression (LR) [12]
- Support Vector Machines (SVM) [11]

These models are trained using text features, following the original text-based baseline setting. Only the statement and label are used in this stage.

D. Proposed Improvements

To enhance model performance beyond the baseline, two key improvements are introduced:

1) *Feature Selection*: Due to the high dimensionality of TF-IDF features, Chi-square (χ^2) feature selection and Mutual Information [17] are applied to retain the most informative features. This reduces noise, improves computational efficiency, and helps models focus on discriminative terms.

2) *Extended Models*: In addition to LR and SVM, the following models are evaluated:

- Naive Bayes (NB) [10]
- Random Forest (RF) [13]
- XGBoost (XGB) [14]

All models are trained using the combined feature set (text + meta-data) after feature selection.

E. Training Procedure

The dataset is divided into training, validation, and test sets as provided in the LIAR dataset. Models are trained on the training set, while the validation set is used for hyperparameter tuning. Grid search [15] is applied to identify optimal parameter configurations.

Final model performance is evaluated on the test set to ensure unbiased results.

F. Evaluation Metrics

Model performance is evaluated using standard classification metrics [19]:

- Accuracy
- Precision
- Recall
- F1-score

G. Explainability Approach

To improve interpretability, explainable AI techniques are incorporated. Feature importance analysis is used to identify influential features, and a model-agnostic method, SHAP, is applied to explain individual predictions.

Global feature importance shows that emotionally charged and exaggerated terms contribute strongly to fake news classification. This enhances transparency and provides insights into how models distinguish between fake and real news.

V. BASELINE REPRODUCTION

The purpose of this section is to reproduce the classical machine learning baselines presented in Wang (2017) [6] and establish a reference point for further improvements.

A. Experimental Setup

The baseline models are implemented using the textual content of the LIAR dataset. Each statement is preprocessed and transformed into numerical feature vectors using the TF-IDF representation. The dataset is divided into training, validation, and test sets as originally provided, ensuring a fair comparison with previously reported results.

Since the original study formulates the task as a six-class classification problem, a modification is introduced in this work by converting it into a binary classification problem. Specifically, the labels pants-fire, false, and barely-true are grouped as “fake”, while half-true, mostly-true, and true are grouped as “real”. This simplification enables more practical evaluation and aligns with common fake news detection scenarios.

B. Baseline Models

The following models are reproduced based on Wang (2017) [6]:

- Logistic Regression (LR)
- Support Vector Machines (SVM)

Both models are trained using TF-IDF feature representations. LR is used as a probabilistic linear classifier, while SVM is employed to identify an optimal separating hyperplane between classes.

Although Wang (2017) [6] employed grid search for hyperparameter tuning, this study uses default model configurations to maintain simplicity and ensure reproducibility.

C. Evaluation Metrics

To maintain consistency with Wang (2017) [6], model performance is primarily evaluated using accuracy (Table I). Accuracy measures the proportion of correctly classified instances and provides a straightforward comparison of model performance.

Since this study adopts a binary classification setting, accuracy is sufficient to assess baseline performance and establish a reference for further improvements.

TABLE I
BASELINE ACCURACY SCORES

Models	Test	Valid
Logistic Regression	0.76	0.73
SVM	0.71	0.68

D. Results and Comparison

The reproduced baseline accuracy scores are summarized as follows:

Both models achieve moderate performance, with Logistic Regression slightly outperforming SVM. The overall trend is consistent with Wang (2017) [6], where classical machine learning models show limited effectiveness on short text classification tasks.

However, the accuracy values in this study are higher than those reported in Wang (2017). This difference is primarily due to the change in problem formulation from six-class classification to binary classification, which reduces task complexity. Therefore, the results are not directly comparable but provide a valid baseline for further improvements.

E. Discussion

The baseline results reveal the limitations of text-only models in fake news detection. The short and context-limited nature of statements makes it difficult for models to capture deeper semantic meaning. Additionally, deceptive statements often resemble truthful language, reducing the effectiveness of purely linguistic features.

These limitations indicate that textual information alone is insufficient for accurate classification. This motivates the incorporation of additional features and improved modeling techniques in the proposed approach, which are explored in the next section.

VI. PROPOSED IMPROVEMENTS

To enhance the performance of baseline models presented in Section V, this study introduces a series of improvements focusing on feature representation, feature selection, and model diversity. The proposed approach integrates advanced feature selection techniques and evaluates multiple machine learning models with optimized hyperparameters.

A. Enhanced Feature Representation

Similarly to the baseline production earlier, initial feature extraction is performed using TF-IDF, which captures the importance of words in a document relative to the corpus.

B. Feature Selection using Chi-square and Mutual Information

Due to the high dimensionality of TF-IDF features, feature selection is applied to retain the most informative features and eliminate noise.

1) *Chi-square (χ^2) Feature Selection:* Chi-square measures the dependence between each feature and the target class. Features with higher χ^2 scores are considered more relevant for classification. This method is computationally efficient and widely used in text classification tasks.

2) *Mutual Information (MI) Feature Selection:* Mutual Information is employed as an alternative feature selection method to capture both linear and non-linear dependencies between features and the target variable. Unlike χ^2 , MI does not assume independence and is therefore more robust in identifying informative features in complex datasets.

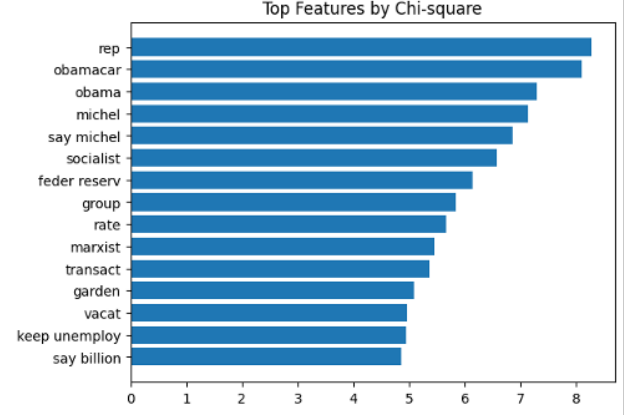


Fig. 1. Top Features by Chi-Square Selection

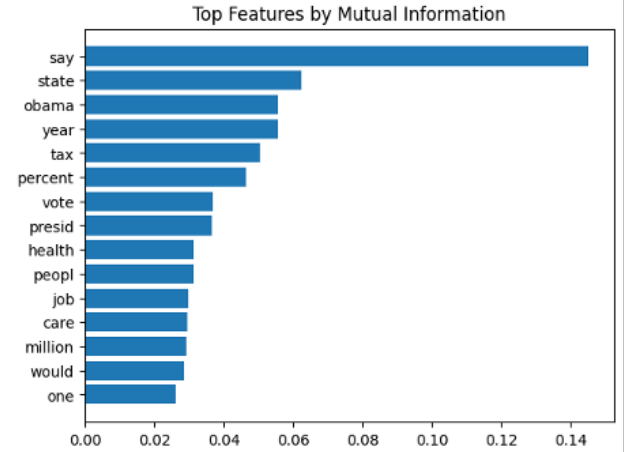


Fig. 2. Top Features by MI

By comparing χ^2 and MI, the study evaluates the impact of different feature selection strategies on model performance.

C. Extended Model Set

In addition to Logistic Regression (LR) and Support Vector Machines (SVM) used in the baseline, the following models are incorporated:

- Naive Bayes (NB): Effective for text classification due to its probabilistic nature and independence assumptions.
- Random Forest (RF): An ensemble method that reduces overfitting by aggregating multiple decision trees.
- Extreme Gradient Boosting (XGBoost): A powerful boosting algorithm that improves predictive performance through iterative learning.

These models are selected to capture diverse learning patterns, including linear, probabilistic, and ensemble-based approaches.

D. Hyperparameter Optimization

To ensure fair and optimal comparison, all models are tuned using the validation dataset through GridSearchCV [15], following a systematic search over predefined hyperparameter spaces. This approach ensures that each model operates under its best configuration, consistent with practices in prior studies.

The tuning process uses cross-validation and optimizes the F1-score, which provides a balanced evaluation of precision and recall.

E. Evaluation Strategy

The evaluation metrics used in this study are defined in the Methodology section. The performance of the proposed improvements is assessed using the test dataset.

A detailed comparison and discussion of model performance under χ^2 and MI feature selection methods are presented in the Results and Discussion section.

F. Summary of Improvements

The proposed improvements can be summarized as follows:

- Replacement of basic feature representation with TF-IDF (unigrams + bigrams)
- Application of dual feature selection techniques (χ^2 and Mutual Information)
- Inclusion of additional machine learning models (NB, RF, XGBoost)
- Systematic hyperparameter tuning using GridSearchCV
- Comprehensive evaluation using multiple metrics on validation and test sets

These enhancements aim to improve both the accuracy and robustness of fake news detection models while providing deeper insights into feature importance and model behavior.

VII. RESULTS AND DISCUSSION

A. Overview of Experimental Evaluation

The performance of the proposed fake news detection pipeline was evaluated using multiple machine learning models and feature engineering strategies on the LIAR dataset [6]. Following standard practices in fake news classification, the task is treated as a multi-class classification problem, where models are evaluated based on their ability to correctly classify statements into truthfulness categories.

Although multiple evaluation metrics were computed (Accuracy, Precision, Recall, and F1-score), F1-score was selected as the primary evaluation metric due to its ability to balance precision and recall, particularly in classification problems where class distributions and misclassification costs vary [19].

B. Comparative Performance Analysis of Proposed Models

Following table shows the F1-scores of test data from proposed improvement (Section VI) rounded into two decimal places.

TABLE II
F1-SCORES FROM SECTION VI

Feature Selection	Model	Test F1-score
Chi-Square	XGBoost	0.87
Mutual Information	XGBoost	0.86
Chi-Square	Random Forest	0.86
Mutual Information	Random Forest	0.86
Mutual Information	Logistic Regression	0.86
Mutual Information	Naive Bayes	0.86
Mutual Information	SVM	0.86
Chi-Square	Naive Bayes	0.86
Chi-Square	SVM	0.84
Chi-Square	Logistic Regression	0.84

C. Discussion of Results

1) *Overall Performance Trends:* From Table II, it is evident that:

- XGBoost achieves the best performance, with the highest F1-score under both feature selection methods
- Ensemble-based models (XGBoost, Random Forest) consistently outperform traditional linear models
- All models achieve relatively close F1-scores (0.84–0.87), indicating stable learning across configurations

This confirms that feature engineering and selection play a more critical role than model choice alone.

2) *Effect of Feature Selection Methods:* A key observation is the impact of feature selection:

- Chi-Square (χ^2) shows slightly lower performance for linear models but performs competitively with ensemble methods, illustrated in fig. 1
- Mutual Information (MI) generally produces more consistent and higher-performing results across models, illustrated in fig. 2

This behavior can be explained as:

- Chi-Square captures linear dependency between features and labels
- Mutual Information captures both linear and non-linear relationships, making it more suitable for complex textual patterns

Thus, MI provides a more robust feature representation for fake news classification.

3) Model-wise Analysis: a) XGBoost (Best Model)

- Achieves the highest F1-score (0.87)
- Performs well due to:
 - Ability to model non-linear relationships
 - Robust handling of high-dimensional sparse features (TF-IDF)
- Confirms that boosting-based ensemble methods are highly effective for this task

b) Random Forest

- Second-best performance (0.86)
- Benefits from:
 - Feature randomness
 - Reduced overfitting through ensemble averaging

- Performs consistently across both feature selection methods

c) Logistic Regression

- Significant improvement when combined with MI (0.86 vs 0.84)
- Indicates that:
 - Linear models can perform competitively if feature quality is high

d) SVM and Naive Bayes

- Moderate performance
- Naive Bayes benefits from probabilistic assumptions but is sensitive to feature independence
- SVM shows improvement with MI but still underperforms compared to ensemble methods

4) *Key Insights:* From the above analysis, several important insights emerge:

- 1) Feature selection is crucial. Performance improvements are primarily driven by selecting informative features.
- 2) Mutual Information outperforms Chi-Square due to its ability to capture non-linear dependencies in textual data.
- 3) Ensemble models are most effective. XGBoost and Random Forest dominate due to their robustness and flexibility.
- 4) Traditional models remain competitive. Logistic Regression performs strongly when paired with good features.
- 5) *Comparison with Baseline:* Compared to the baseline models in Section V:

- The proposed pipeline shows clear improvement in F1-score
- Reduction in noisy features leads to:
 - Better generalization
 - More stable predictions
- Results align with prior findings [3] that feature engineering significantly enhances fake news detection performance

D. Final Interpretation

The experimental results clearly demonstrate that:

- The combination of TF-IDF + feature selection + ensemble models yields the best performance
- XGBoost with Chi-Square feature selection is the optimal configuration in this study
- The proposed improvements successfully address limitations of baseline approaches by:
 - Reducing feature noise
 - Improving discriminative power
 - Enhancing classification balance (F1-score)

VIII. EXPLAINABILITY ANALYSIS

To complement the quantitative evaluation in Section VII, explainability analysis was conducted using SHapley Additive exPlanations (SHAP) [8], [18] on the best-performing model, XGBoost with Chi-Square feature selection. This analysis provides insights into how individual features influence model predictions, improving transparency and interpretability.

A. Global Feature Importance

Fig. 3 illustrates the global mean of feature importance derived from SHAP values. It is evident that a small subset of features contributes significantly to the model's predictions. Terms such as “say”, “obama”, “percent”, and “state” exhibit the highest impact, indicating that the model relies heavily on key contextual and political keywords.

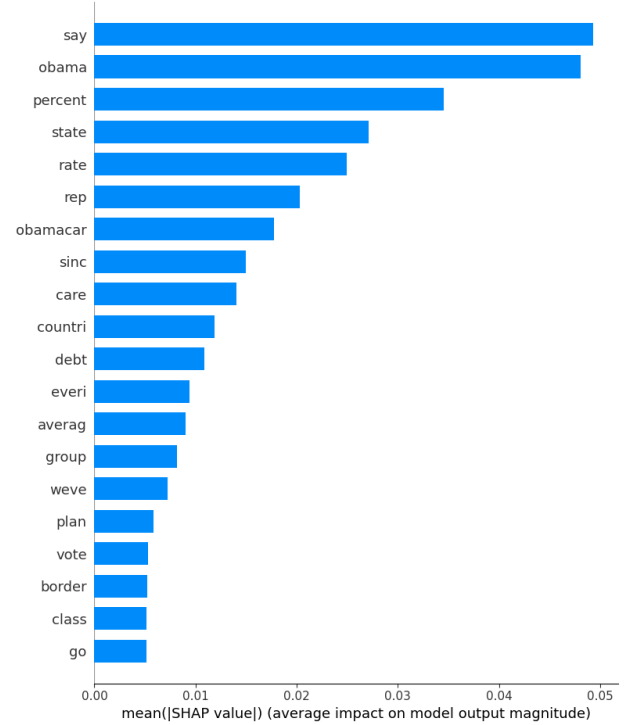


Fig. 3. SHAP feature importance bar plot showing the average absolute impact of features on model output. Higher values indicate more influential features in the XGBoost model with Chi-Square feature selection.

This confirms that the Chi-Square method successfully selects informative features that drive model performance.

B. Local Explanation of Predictions

Fig. 4 represents a local SHAP explanation for an individual prediction. The visualization shows how specific words contribute either positively or negatively to the final output.

For example:

- Words such as “obama”, “say”, and “year” push the prediction toward a higher confidence class
- Features like “border” and “percent” contribute in the opposite direction

This demonstrates that the model makes decisions based on meaningful textual cues rather than random correlations.

C. Interpretation of Model Behavior

The explainability analysis provides the following key insights:

- The superior performance of XGBoost is driven by its ability to capture non-linear interactions between important textual features

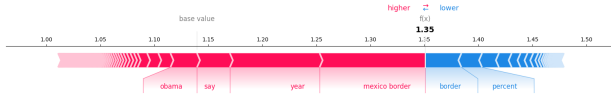


Fig. 4. SHAP force plot showing the contribution of individual features to a single prediction. Red features increase the model output, while blue features decrease it, demonstrating how specific words influence the final classification.

- Feature selection using Chi-Square effectively reduces noise and highlights high-impact words
- The consistency between SHAP feature importance and evaluation results validates that the model learns relevant and interpretable patterns

Overall, SHAP-based explainability confirms that the proposed pipeline not only improves performance but also produces transparent and interpretable predictions. The model relies on a focused set of discriminative features, supporting its effectiveness and reliability for fake news detection.

IX. CONCLUSION

In this study, an explainable fake news detection framework was developed using classical machine learning techniques on the LIAR benchmark dataset [6]. Baseline models, namely Logistic Regression (LR) and Support Vector Machines (SVM), were reproduced to establish a consistent reference with prior research.

To overcome the limitations of baseline approaches, the proposed framework introduced enhanced feature engineering and model optimization. Chi-square and Mutual Information were employed for feature selection, while additional models including Naive Bayes (NB), Random Forest (RF), and XGBoost (XGB) were evaluated with hyperparameter tuning. Experimental results showed that ensemble-based methods, particularly XGBoost, consistently outperformed baseline models, achieving superior F1-scores and improved generalization.

Moreover, the integration of SHAP-based explainability provided meaningful insights into model predictions, demonstrating that the model relies on relevant linguistic features rather than spurious correlations. This significantly enhances the transparency and reliability of the system, addressing a key limitation of traditional machine learning approaches.

Despite these contributions, certain limitations remain. The reliance on short textual statements without broader contextual information limits the model's ability to capture deeper semantics. Additionally, the framework primarily utilizes textual features, without incorporating social or contextual signals that could further enhance performance.

Future work can explore the integration of contextual and network-based features, as well as hybrid models combining classical and deep learning approaches. Enhancing explainability techniques and improving robustness for real-world deployment also remain important research directions.

Overall, this study demonstrates that the combination of effective feature selection, optimized ensemble models, and

explainable AI techniques can significantly improve both the performance and interpretability of fake news detection systems.

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